

Hybrid Telecom Stream Processing Framework

MPI-Based Multi-DES Communication for Distributed Telecom Packet Processing

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Telecom Streams Have Outgrown Batch Processing



Data Explosion

5G and LTE networks generate continuous, high-velocity packet streams that defy traditional batch processing.



Real-Time Demands

Providers need sub-millisecond analytics for anomaly detection, billing, and network optimisation.



Scalability Necessity

Handling large-scale packet volumes demands distributed, parallel computing to avoid bottlenecks.

THE SOLUTION

The Hybrid Telecom Stream Processing Framework integrates distributed Data Extraction Servers (DES) with a parallelised MPI-based backend for reliable, high-throughput packet handling — from ingestion to storage, analytics, and monitoring.

Telecom Infrastructure Buckles Under Peak Traffic



Distributed Complexity

Synchronising traffic across multiple, concurrently active Data Extraction Servers.



Communication Latency

Conventional messaging protocols struggle to keep pace with high-speed packet ingestion.



Concurrency Overhead

Shared resources must be managed without compromising ordering or throughput.



Resource Constraints

Need for continuous runtime monitoring and non-blocking, reliable storage.

Monolithic architectures fail under bursty traffic — resulting in packet drops and degraded QoS.

What the Framework Set Out to Achieve



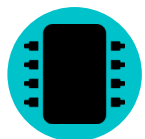
Communication Architecture

Develop an MPI-based framework for reliable, low-latency communication between distributed nodes.



Scalable Extraction

Support multiple, concurrently active DES clients without redesigning the server's core logic.



Fine-Grained Parallelism

Use POSIX Threads for concurrent, high-throughput packet processing within each node.



Data Reliability

Integrate Berkeley DB for persistent, high-speed storage of every processed packet.

Integrated System Architecture



A decoupled producer-consumer pipeline: MPI decouples ingestion from processing, so a burst of incoming DES traffic never blocks the communication layer.

Modular

Independently Testable

Composable

Core Technology Stack



C

Language



POSIX Threads

Parallelism



MPI (OpenMPI)

Distributed Comms



TCP Sockets

Early-Phase Networking



Berkeley DB

Storage



Mutex / CondVar

Synchronisation



Queue & Graph

Data Structures



GNU Make

Build System



Ubuntu Linux

Operating System



Git & GitHub

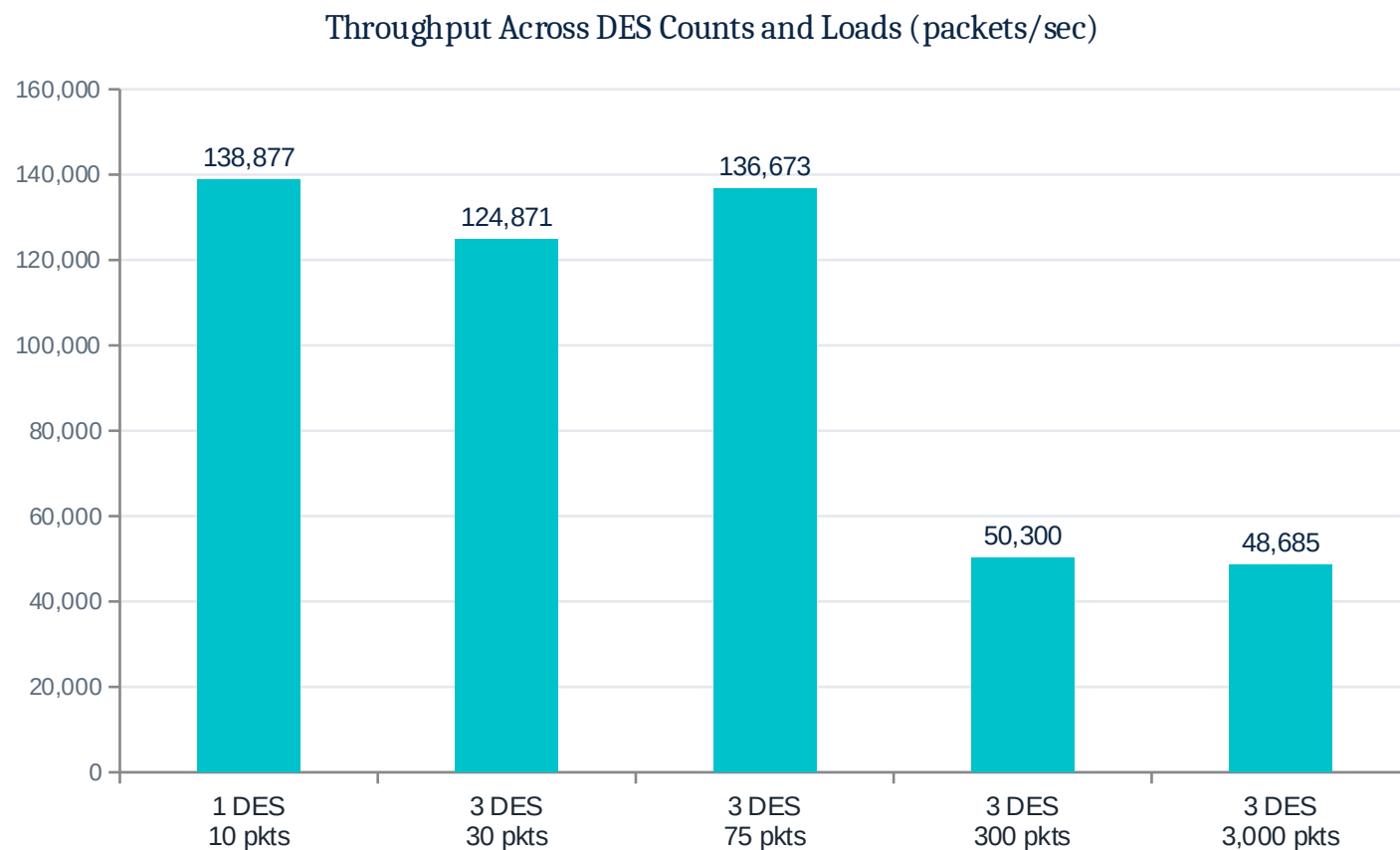
Version Control

Implementation Milestones



Each phase — independently designed, implemented, tested, and validated — before progressing to the next.

Performance & Throughput Under Load



KEY FINDINGS

- Stable 48.7K–50.3K pkts/sec sustained across a 10x jump in load (300 → 3,000 pkts)
- Under 4% throughput variance despite the tenfold increase in packet volume
- Zero packet loss, deadlocks, or race conditions at every scale tested

Validation & Quality Assurance



Integrity

FIFO packet sequencing maintained throughout the entire stream pipeline, in every configuration tested.



Synchronisation

Extensive stress testing across five load levels — no deadlocks or race conditions observed.



Functional Coverage

Producer/consumer modules and MPI-DES handshake fully validated under synthetic traffic generators.

5 / 5

Test Configs Passed

0

Packets Lost

3,000

Peak Packets Validated

Future Research & Scaling



Cloud Integration

Transition to a Kubernetes-managed, containerised deployment.



Predictive Analytics

AI/ML models for proactive traffic prediction and load balancing.



Fault Tolerance

Checkpointing mechanisms to withstand node failure at scale.



Visualization

GUI-based analytics dashboard for real-time network state.

SUMMARY

A Production-Ready Foundation

The Hybrid Telecom Stream Processing Framework delivers a high-performance, scalable, and reliable architecture for modern telecom packet analytics — decoupling extraction from processing via MPI and POSIX Threads to solve the latency and concurrency challenges of massive data streams.

48.7K+

pkts/sec sustained throughput

3×

concurrent DES clients validated

0

packets lost across all tests

Thank You

Questions & Discussion

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